Muhammad Arham

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PROFESSIONAL SUMMARY

Recent Computer Science graduate with skills in HTML5, CSS3, JavaScript, React.js, Tailwind CSS, Node.js, Flask, and SQL. Eager to contribute, learn, and grow as a Frontend Developer and Software Engineer.

SKILLS

Programming Languages :	JavaScript (ES6+), Python, C++
Frontend :	React.js, React Router, Redux, Context API, HTML5, CSS3, Responsive Design, Cross-Browser Compatibility
UI Frameworks :	Tailwind CSS, Bootstrap, Material UI
Backend :	Node.js, Express.js, Flask, REST APIs (Axios)
Databases :	SQL, MongoDB, Firebase
Tools & Platforms :	Git, GitHub, npm, Vite, VS Code, Netlify, Vercel
ML / Image Processing :	OpenCV, TensorFlow
Soft Skills :	Problem Solving, Communication, Teamwork, Adaptability, Quick Learning, Time Management

WORK EXPERIENCE

Web Development Intern (3 months) | Internship — Mar 2025 - May 2025

VirtuoSoft Software House — *Karachi, Pakistan*

- **Built** responsive, **cross-browser** web pages using **HTML5, CSS3,** and **JavaScript**, converting design mockups into functional, pixel-accurate interfaces that render consistently across modern browsers.
- **Developed** and maintained **reusable front-end components**, improving layout consistency, reusability, and usability across devices.
- **Debugged** and tested application features to identify and resolve UI issues prior to deployment, reducing post-release defects and improving overall product quality.
- **Collaborated** with a cross-functional team of developers and designers to translate project requirements into working web pages.
- **Refined** page navigation and interface responsiveness to improve the overall user experience, and supported ongoing website development projects through coding, testing, and technical documentation.

Technologies / Skills Used : *HTML5, CSS3, JavaScript*

PROJECTS

Crypto Decoder — Trading Signals Platform |

- **Solved** signal groups stuck on chat screenshots and sheets with one **full-stack platform** where admins post, members see signals live.
- **Built** instant alerts with **WebSockets** and a live chart, so members see entry, stop loss and targets the moment they are posted.
- **Added** a **paper-trading** terminal with virtual money, secure login with 2FA, and one-click **Excel export** of every signal and trade.

Technologies / Tools Used : *React, Vite, Node.js, Express, SQLite, WebSockets, Tailwind CSS*

Fingerprint Re-Touch ID — Final Year Project | — 2026

- **Engineered end-to-end data pipelines** and trained **deep learning models** including **CNN, U-Net, and CycleGAN** for automated image restoration and biometric feature recovery.
- **Designed a scalable model integration strategy** to enhance biometric recognition efficiency for large-scale authentication systems, such as NADRA and Banking Systems.
- **Developed** a machine learning-based system to enhance fingerprint images for biometric recognition, applying **OpenCV** and **TensorFlow** image-processing techniques to improve ridge clarity and reduce noise.
- **Processed** and prepared image datasets for model training and testing, directly improving fingerprint recognition accuracy.
- **Built** a React front end integrated via **REST API (Axios)** with a Python/Flask backend to demonstrate the enhancement pipeline and results.

Technologies / Tools Used : *Python, Google Colab, CNN, U-Net, CycleGAN, OpenCV, TensorFlow, React, Flask, Axios*

Web Development Projects (Portfolio) |

- **Designed and built** multiple **responsive web applications** using HTML5, CSS3, JavaScript, and React, ensuring consistent functionality across all devices.
- **Implemented** backend logic and data handling using **Flask** and **SQL** to support core application features.
- **Focused** on clean UI architecture and responsive design principles to deliver a smooth user experience across all screen sizes.

Technologies / Tools Used : *HTML5, CSS3, JavaScript, React, Flask, SQL*

EDUCATION

NED University of Engineering and Technology | Karachi — 2022 - 2026

Bachelor of Science in Computer Science — *CGPA : 3.00/4.00*

Chairman's Medal for **Best Paper Project**: "Fingerprint Enhancement Using Machine Learning", recognized at an International Conference for research excellence.

S.A.L College — 2022

Pre-Engineering — *Grade : A1*

HSC — *Mirpurkhas* Board

Laraib Public School — 2020

Science — *Grade : A*

SSC — *Sukkur* Board

AWARDS & ACHIEVEMENTS

- **Best Research Paper Award – 1st Position** – Received for the final year project research paper, **"Fingerprint Enhancement Using ML."** |